Arista Access Node KOR-ACC-79 QoS Congestion Event Resolution SOP

Node: KOR-ACC-79 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9473

Description

The failure of Node ID KOR-ACC-79 at the Koramangala Datacenter site is attributed to a QoS congestion event, specifically microbursts and interface tail drops, leading to QoS queue starvation. This phenomenon is a direct consequence of the Access role's inability to adequately manage the bursty traffic patterns, which overwhelm the interface buffers and subsequently cause the router to discard packets. This, in turn, triggers the ALARM_9473 major alarm. The root cause lies in the misconfiguration of the QoS policies, which fail to provide sufficient policing and shaping mechanisms to mitigate the effects of the bursty traffic. Furthermore, the Arista access node's hardware capabilities, such as the CPU and memory resources, may be insufficient to handle the high-traffic loads, exacerbating the congestion issue.

Pre-requisite Checks

- Physical presence at the Koramangala Datacenter site to access the affected node
- Verification of redundant link status to ensure seamless failover in case of a node failure
- Review of recent configuration changes to identify potential causes of the QoS congestion event

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate CPU and memory resources on the access node	Medium
High-traffic loads exceeding the interface buffer capacities	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware cpu, Expected Output: CPU utilization percentage, Description: This command is used to identify any CPU bottlenecks that may be contributing to the QoS congestion event. High CPU utilization can indicate that the access node is struggling to process the high-traffic loads, leading to packet discard and QoS queue starvation.

Command: show interface statistics, Expected Output: Interface statistics, including packet discard and error counts, Description: This command provides detailed statistics about the interface, including packet discard and error counts. By analyzing these statistics, we can determine if the interface is experiencing congestion and if packet discard is occurring due to buffer overflow.
```

```
Command: show qos policy, Expected Output: QoS policy configuration, Description: This command is used to verify the QoS policy configuration on the access node. We can check if the policy is correctly configured to police and shape the traffic, and if the policy is correctly applied to the interface experiencing congestion.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: configure terminal set qos police 1000 1000 set qos shape 1000 1000 end, Description: We need to review and adjust the QoS policy configuration to ensure that it is correctly policing and shaping the traffic. We can use the 'set qos police' and 'set qos shape' commands to configure the policy and apply it to the interface experiencing congestion.

Step 2: Step: Increase CPU and memory resources on the access node, Command: configure terminal upgrade cpu 2 upgrade memory 16GB end, Description: If the CPU and memory resources on the access node are insufficient to handle the high-traffic loads, we need to upgrade the resources to ensure that the node can process the traffic without congestion. We can use the 'upgrade cpu' and 'upgrade memory' commands to increase the CPU and memory resources.

Step 3: Step: Implement traffic shaping and policing on the interface, Command: configure terminal interface gigabitethernet 0/0 service-policy input shape 1000 1000 exit exit, Description: We need to implement traffic shaping and policing on the interface experiencing congestion to ensure that the traffic is correctly managed and packet discard is avoided. We can use the 'service-policy input shape' command to apply the policy to the interface.

Post-Check Verification

```
Command: show hardware cpu, Expected Output: CPU utilization percentage < 50%, Description: We need to verify that the CPU utilization has decreased and is below the threshold of 50% after implementing the resolution steps.

Command: show interface statistics, Expected Output: Interface statistics, including packet discard and error counts, show significant reduction in discard and error counts, Description: We need to verify that the interface statistics show a significant reduction in packet discard and error counts after implementing the resolution steps.

Command: show qos policy, Expected Output: QoS policy configuration, showing correct policing and shaping, Description: We need to verify that the QoS policy configuration shows correct policing and shaping to ensure that the traffic is correctly managed.
```

Roll-back Procedures

- Step: Revert QoS policy configuration to previous version, Command: configure terminal rollback 1 end, Description: If the fix fails, we need to revert the QoS policy configuration to the previous version to ensure that the original configuration is restored.

- Step: Downgrade CPU and memory resources on the access node, Command: configure terminal downgrade cpu 1 downgrade memory 8GB end, Description: If the fix fails, we need to downgrade the CPU and memory resources on the access node to their original configuration to ensure that the node's resources are reverted to their original state.

- Step: Remove traffic shaping and policing on the interface, Command: configure terminal interface gigabitethernet 0/0 no service-policy input shape 1000 1000 exit exit, Description: If the fix fails, we need to remove the traffic shaping and policing on the interface to ensure that the original configuration is restored.